

BONN ROLEBER, GermanyWMO: **105190**Lat: **50.7350N**Lon: **7.1931E**Elev: **526**StdP: **14.42**Time Zone: **1.00 (EUC)**Period: **01-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.1	21.8	11.1	9.9	22.2	15.1	12.1	25.2	20.6	47.4	18.6	44.5	4.6	110	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	84.6	67.6	81.2	66.4	78.0	65.0	69.7	80.6	68.2	77.9	66.6	75.4	6.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	99.1	73.3	64.6	93.1	71.5	63.2	88.6	70.1	34.0	80.3	32.7	78.6	31.5	75.4	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.3	14.3	12.7	DB	13.3	91.7	5.7	2.5	9.2	93.6	5.9	95.1	2.7	96.5	-1.5	98.4
			WB	12.5	72.5	5.7	1.5	8.4	73.6	5.1	74.5	1.8	75.4	-2.3	76.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	36.2	37.2	43.4	50.7	56.7	62.3	65.8	64.9	58.8	51.8	44.4	37.0	
	DBStd	12.55	8.36	7.91	7.80	7.27	6.57	5.94	5.83	4.98	5.62	6.78	6.23	7.32	
	HDD50	1770	429	362	229	78	17	1	0	0	2	62	186	404	
	HDD65	5378	891	779	670	429	267	121	62	63	201	409	618	868	
	CDD50	2078	3	3	24	100	225	369	491	460	266	118	18	1	
	CDD65	213	0	0	0	1	10	39	88	59	15	1	0	0	
	CDH74	1751	0	0	0	28	92	317	745	449	118	2	0	0	
	CDH80	423	0	0	0	0	17	58	216	117	15	0	0	0	
Wind	WSAvg	6.3	7.3	7.1	6.8	6.3	5.9	5.6	5.6	5.6	5.6	6.3	6.7	7.1	
Precipitation	PrecAvg	27.7	1.9	1.9	1.9	1.7	2.4	3.0	3.0	3.0	2.4	2.2	2.2	2.2	
	PrecMax	32.2	4.4	4.3	3.8	3.4	4.6	5.5	7.4	6.1	5.3	4.8	4.3	5.0	
	PrecMin	21.7	0.3	0.3	0.2	0.0	0.8	1.2	1.2	0.9	0.7	1.1	0.0	0.7	
	PrecStd	2.9	1.0	0.9	0.9	0.8	1.0	1.1	1.4	1.7	1.3	0.9	0.8	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	57.4	66.5	77.9	82.1	86.0	90.5	89.5	82.3	71.8	61.5	55.2	
		MCWB	51.3	50.0	52.8	58.4	66.0	69.3	68.8	69.3	65.8	62.8	54.4	51.7	
	2%	DB	53.3	53.7	61.9	72.8	76.0	81.2	85.5	82.8	77.4	67.4	58.2	51.8	
		MCWB	49.4	48.9	51.4	56.6	61.5	67.1	67.7	67.1	65.0	59.8	52.9	48.9	
	5%	DB	50.7	51.0	58.4	68.0	72.6	77.7	81.7	79.0	73.2	64.2	55.9	49.3	
		MCWB	47.7	46.6	50.4	54.6	59.6	64.6	66.5	66.3	63.0	58.2	51.7	46.6	
	10%	DB	47.8	48.2	54.9	63.8	69.0	74.2	78.1	75.2	69.1	61.5	53.0	47.0	
		MCWB	45.1	44.5	48.4	53.2	58.1	63.4	65.5	64.2	60.3	56.7	49.8	44.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	51.3	55.4	60.2	68.3	71.7	71.9	70.6	68.3	63.4	55.5	52.2
			MCDB	54.9	54.4	61.6	72.3	79.2	82.2	82.7	81.5	78.3	69.6	59.0	54.6
2%		WB	49.8	49.3	53.2	57.9	64.7	68.7	69.9	68.9	66.0	60.8	53.6	49.1	
		MCDB	52.5	52.9	59.0	69.7	72.4	78.3	80.6	79.8	75.2	65.9	57.3	51.1	
5%		WB	48.0	47.0	51.4	56.1	61.7	66.6	68.3	67.3	63.7	59.1	52.1	46.9	
		MCDB	50.6	50.4	57.0	65.6	68.9	74.5	77.9	76.7	71.2	63.5	55.5	48.9	
10%		WB	45.1	44.8	49.1	54.2	59.2	64.6	66.6	65.5	61.7	57.1	50.1	44.7	
		MCDB	47.4	47.8	54.2	62.7	66.2	71.7	75.7	73.1	67.9	60.7	52.9	46.9	
Mean Daily Temperature Range		5% DB	MDBR	8.0	9.3	13.3	16.9	16.5	17.1	17.0	16.1	15.5	12.8	9.0	7.6
			MCDBR	9.3	11.7	18.8	24.3	22.9	23.6	24.5	23.4	22.2	16.4	10.8	8.8
	MCWBR		7.7	7.8	10.5	11.7	11.0	11.1	9.8	10.5	11.4	9.6	7.3	7.6	
	5% WB	MCDBR	9.6	9.4	15.8	20.8	20.0	20.1	21.2	20.0	19.9	14.8	10.2	8.7	
		MCWBR	8.4	7.2	9.1	10.4	10.6	10.6	9.4	9.7	10.8	9.4	7.4	7.9	
Clear-Sky Solar Irradiance	taub	0.313	0.338	0.369	0.403	0.409	0.411	0.416	0.410	0.380	0.358	0.334	0.311		
	taud	2.432	2.382	2.312	2.253	2.270	2.279	2.264	2.309	2.373	2.429	2.429	2.414		
	Ebn at Noon	228	251	262	265	269	269	265	260	256	240	216	209		
	Edh at Noon	19	26	33	39	40	40	40	37	31	25	19	17		
All-Sky Solar Radiation	RadAvg	238	456	814	1301	1538	1686	1597	1369	1012	591	275	182		
	RadStd	37	87	124	151	144	158	191	133	126	59	36	26		

Historical Trends

		DBAvg	Heating		Cooling		
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